ASSET-LIGHT ENGINEERING SERVICES

Industrial Rooftop Solar EPC & Contracting (AISEC)

A strategic blueprint for launching a high-margin, asset-light solar engineering, procurement, and contracting (EPC) firm targeting GIDC factory roofs in Gujarat.

PROJECT PROMOTER

Lead Entrepreneurial Group

PRIMARY OPERATIONAL HUB

Changodar Engineering Corridor (Ahmedabad)

PHASE 1 CAPITAL PLAN

₹15 Lakhs (Working Capital & Software)

TARGET CAPITAL ASSET RATING

Asset-Light (Outsourced EPC Labor)

Table of Contents

1. Executive Summary

As Gujarat's industrial power tariffs cross **₹7.80 per unit** (exclusive of fuel surcharges and electricity duty), manufacturing companies in GIDC zones are looking to offset their overheads. The **Gujarat Solar Policy** allows 100% net-metering on rooftop solar installations up to contract demands, creating a highly viable investment case for factory owners who can recover installation costs in under 3.5 years.

The proposed **Industrial Rooftop Solar EPC & Contracting (AISEC)** is an engineering services firm structured to address this market. Deploying a lean, asset-light operational model, the firm manages customer acquisition, GEDA (Gujarat Energy Development Agency) registrations, structural roof engineering, and equipment sourcing. Physical installation work is outsourced to certified Class-A subcontractors.

AISEC operates on a milestones-funded payment schedule, requiring a low initial starting capital of ₹15 Lakhs for working capital. The company partners with local solar panel distributors to secure equipment on credit terms backed by customer advance payments, ensuring positive cash flow from Day 1 and avoiding downstream capital lock-ins.

Core Business Intent

By establishing a specialized engineering firm focused on factory solar installations in the industrial zones of Sanand and Changodar, AISEC secures high-margin project design fees while outsourcing execution risks.

2. Future Potential of the Industry

The commercial and industrial (C&I) rooftop solar sector in India is expanding rapidly, supported by state net-metering benefits and clean energy policies.

Additionally, rising ESG reporting requirements are driving corporate manufacturing entities to secure green power offsets, establishing a highly receptive customer base for rooftop solar installations.

2.1 Industrial Tariff & GERC Net-Metering Rules

The GERC (Gujarat Electricity Regulatory Commission) net metering policy sets clear boundaries for C&I installations:

Capacity Category	Net/Gross Metering Rules	Target Consumers	Surplus Compensation Rates
1 kW to 1 MW Capacity	Net Metering permitted up to contract demand limits.	Mid-sized manufacturing units, cold storages, GIDC sheds.	Compensated at GERC-determined rates (linked to average power purchase cost).
10 kW to 1 MW Option	Gross Metering permitted as an alternative framework.	Commercial buildings, landlords with unutilized roof space.	Fixed feed-in-tariff rates for all generated power injected to grid.
Above 1 MW Capacity	Open Access Captive Solar required; net metering not available.	Heavy chemical plants, large automotive assembly hubs.	Subject to cross-subsidy surcharges and wheeling charges.

2.2 Dholera & Olympic Infrastructure Green Mandates

The construction of the **Tata-ASML semiconductor fab in Dholera** directly drives local renewable energy demand. High-technology manufacturing plants require stable, zero-carbon energy sources to meet international supply-chain sustainability standards.

Key growth drivers include:

- **Green Supply Chains:** Component manufacturers supplying the Dholera fab are mandated to offset their energy footprints using renewable sources. This drives rooftop solar demand among secondary manufacturing units in GIDC zones.

- **Dholera Smart Grid Interconnects:** The DSIR smart grid supports bidirectionally balanced power distribution, making solar installations in the region highly viable.
- **Ahmedabad Olympics 2036:** The bid for the 2036 Olympic Games requires all sporting venues, athlete housing, and administrative structures to operate on **100% renewable energy**. This opens opportunities for engineering firms to secure commercial green-energy installation and consulting contracts.

3. Siting & Sourcing Hub Strategy

An engineering services firm requires close proximity to major industrial zones for client acquisition:

Proposed Siting Zone	Suitability Rating	Engineering & Sales Pros	Overheads & Office Costs	Recommendation
Changodar Corridor (Ahmedabad Outskirts) Office Rent: ₹15,000 - ₹20,000 / mo	**9.6 / 10** 🏆 Recommended Siting	• Direct proximity to over 2,500 industrial manufacturing sheds. • Easy transport access to Sanand and Bavla zones. • Lower local office overheads.	• Further from the GEDA central office in Gandhinagar (~45 km).	**Winner for Phase 1.** Placing our sales and design team inside the target industrial zone helps accelerate client acquisition.
Gandhinagar Corporate Zone Office Rent: ₹35,000 - ₹45,000 / mo	**7.5 / 10** ⭐ ⭐ Average Siting	• Close to government approval authorities (GEDA, CEIG). • Premium corporate branding.	• High rental burn for sales teams. • Further from the target manufacturing zones in South Ahmedabad.	**Suitable for Phase 2** corporate expansions.

4. National & Local Competitors

The solar installation sector features major utility companies and local installers:

- **Tata Power Solar & Adani Solar (National Giants):**
  Model: Focus on large, utility-scale ground-mount installations and large corporate rooftops.
  AISEC Counter-Strategy: We target mid-sized industrial rooftops (50 kW to 300 kW), offering customized engineering designs and faster local approvals.
- **Local Class-A Electrical Contractors:**
  Model: Focus on general electrical maintenance and wiring work, offering basic solar installations as an add-on service.
  AISEC Counter-Strategy: We focus entirely on solar engineering and optimization (using PVsyst simulation software), helping clients maximize yields and secure faster GEDA approvals.

5. Legal Compliance Matrix

Regulatory Step	Issuing Authority	Setup Cost	Timeline	Role in Project Lifecycle
Class-A Electrical License	CEIG Gujarat	₹50,000	4 Months	Required to sign off on high-voltage electrical installations and grid syncs. Can be outsourced to a certified partner.
GEDA Channel Partner Status	Gujarat Energy Development Agency	₹25,000	2 Months	Allows registration of consumer solar projects for net-metering benefits.
DISCOM Net-Metering Feasibility	PGVCL / UGVCL (GUVNL Portal)	₹10,000 (per project)	1 Month	Form-A submission to verify grid transformer capacity before structural layouts.
Structural Roof Load Certificate	Chartered Civil Engineer	₹15,000 (per project)	1 Week	Verifies that factory roofs can safely support the solar structure wind-load.

6. Equipment Sourcing Strategy

We manage supply chain costs by partnering with local distributors:

1. **Tier-1 PV Modules:** Source solar panels directly from major manufacturers with factories in Gujarat (such as Waaree in Surat or Adani in Mundra), reducing transport costs.
2. **Inverter Supply Agreements:** Partner with international brands (such as Growatt or Sungrow) to secure direct supply channels and long-term warranty support.
3. **Mounting Structures:** Source hot-dip galvanized mounting structures from local steel fabricators in Kathwada GIDC, matching custom wind-load requirements.

7. Client Sourcing Channels

We focus on industrial customers with high power requirements:

- **Cold Storage Units (Bavla Rural):** Cold storages operate continuously and have large rooftops, making them ideal solar customers.
- **Plastic Injection Molding Factories:** High power consumption makes these factories highly receptive to solar energy savings.
- **Chemical Processing Units:** Partner with chemical units in Ankleshwar and Vatva looking to reduce energy costs and meet green reporting targets.

1

GEDA Onboarding

Register the firm as a GEDA Channel Partner to enable streamlined customer application processing.

2

Design Software

License AutoCAD and PVsyst design software to generate professional yield simulations.

3

Subcontractor Agreements

Partner with certified Class-A electrical installation teams to manage on-site structural works.

4

Supplier Accounts

Establish credit lines with local solar panel and inverter distributors in Ahmedabad.

9. Financial Projections

Phase 1 budget focuses on project design, sales operations, and engineering software setup:

CapEx Budget Component	Cost (INR)	Project Operational Economics (100 kW plant)	Budget (INR)
PVsyst & CAD Design Licenses	₹2,50,000	Equipment Cost (Panels & Inverters)	₹26,00,000
GEDA Single-Window Registration	₹1,50,000	Structure Fabrications (Kathwada)	₹4,00,000
Office Setup & Safety Equipment	₹3,00,000	Outsourced Installation Labor	₹2,50,000
Sales & Marketing Capital	₹4,00,000	GEDA & Grid Liaison Charges	₹1,50,000
Operating Cash Reserves	₹4,00,000	Total Project Cost	**₹34,00,000**
Total Setup CapEx	**₹15,00,000**	**Contract Price to Client**	**₹42,00,000**

Asset-Light Cash Flows

A 100 kW project yields a gross profit of **₹8,00,000**. By structuring project payments (e.g., 30% advance on signing, 50% on material delivery, and 20% on grid synchronization), AISEC finances raw equipment purchases using client advances, keeping our internal capital requirements low.

10. Extended Growth Model (RESCO)

As AISEC establishes its track record, the business model can scale by introducing PPAs:

- **Phase 1: Capex Model (Year 1):** Focus on outright sales where the client finances the installation, generating immediate cash flow.
- **Phase 2: RESCO Model (Year 3):** Partner with private equity funds to build solar assets on client roofs. AISEC retains ownership of the panels and sells energy directly to the factory owner under a 15-year PPA.
- **Phase 3: Micro-Grid Integration (Year 5):** Integrate lithium-ion battery storage systems (ESS) to offer complete off-grid power solutions to industrial clients.

11. Risk Analysis & Mitigation

Identified Business Risk	Impact Level	Defensive Mitigation Strategy
Transformer Capacity Halts	**High**	Submit GEDA Form-A feasibility checks on the GUVNL portal before ordering any project equipment.
Rooftop Structural Failures	**High**	Perform structural analysis and secure load certificates from chartered civil engineers before structural layouts.
Panel Price Spikes	**Medium**	Include raw material escalation clauses in client contracts, locking pricing for 30-day windows.

11.1. Worst-Case Stress-Testing & Liquidation Playbook

Stress Scenario	Immediate Financial Impact	Pre-Planned Mitigation & Escape Route
DISCOM rejects net-metering after installation	Grid sync is blocked; payment milestones of ~₹8 Lakhs are delayed.	File grid synchronization applications under GEDA's single-window portal before starting any structural roof work.
Liquidation Trigger (Stop-Loss)	Unrecovered project debts cross ₹10 Lakhs or monthly sales pipeline drops below 100 kW for 5 consecutive months.	**Capital Preservation Liquidation Protocol:** 1. Halt any new equipment ordering and complete active structural works to secure final payments. 2. Sell active client pipelines and GEDA registrations to partner EPC firms (recover ~₹5 Lakhs). 3. Cancel software licenses and exit the Changodar office lease to preserve ~₹10 Lakhs of investor capital.

12. Technical Design & Panel Selection

Rooftop mounting structures are engineered to withstand local wind loads of up to **150 km/h** (based on IS 875 wind speed specifications for coastal Gujarat areas).

We compare panel configurations based on rooftop space constraints:

Module Chemistry Type	Avg. Efficiency Range	Rooftop Space Requirements	Capital Cost Spread
Monocrystalline PERC (Single Face)	20.8% - 21.6%	65 sq ft per kW	Baseline pricing
Bifacial Dual-Glass Modules	22.2% - 23.0%	55 sq ft per kW	+15% premium (yields 12% extra power output from rear albedo reflection)

Solar panels degrade at average rates of **0.5% to 0.7% annually** over their 25-year performance warranties. Structures are designed to minimize shading losses, maximizing plant performance ratios.

13. EHS & Rooftop Safety Protocols

Rooftop solar installations present fall and electrical hazards. All on-site staff must use certified height safety lifelines, hardhats, and insulated boots.

Key safety guidelines include:

- **DC Arc-Fault Protection:** Install automated DC circuit breakers at the string level to prevent fire risks from loose wiring.
- **Rooftop Walkway Access:** Install dedicated walkways to protect solar modules from cracking during maintenance.

14. Local Installer Networks

We partner with certified local electrical installation teams in Changodar, ensuring all subcontractors carry valid DISH safety certifications. This supports local employment and helps manage project delivery timelines during peak seasons.

15. Strategic Summary & Go/No-Go Investment Verdict

🏆 Investment Verdict: GO (Highly Recommended)

Establishing the **Industrial Rooftop Solar EPC & Contracting (AISEC)** is highly recommended. The combination of rising industrial power tariffs and net-metering policies creates a strong customer value proposition, while the asset-light business model minimizes initial CapEx requirements.

Extended Go-Scale Target Model

Growth Phase	Target Volume (kW/year)	Average Profit Margin	Projected Revenues
Phase 1: Capex EPC	500 kW	15% - 18%	₹2 Crores
Phase 2: RESCO PPA Pools	2,500 kW	20% - 24%	₹10 Crores
Phase 3: Smart Micro-Grids	10,000 kW	25% - 30%	**₹38 Crores**

Liquidation & Cash Flow Safeguards

- **Milestone-backed payments:** Enforce strict payment schedules to ensure project stages are fully funded by client advances.
- **Supplier Credit Lines:** Secure credit terms with panel and inverter distributors to optimize working capital cycles.